INCH-POUND

MIL-S-13303G (AR)
AMENDMENT 4
10 April 1991
SUPERSEEDING
AMENDMENT 3
30 May 1989

MILITARY SPECIFICATION

SIGNALS, SMOKE. GROUND, HAND HELD
PARACHUTE, GREEN SMOKE, M128A1,
RED SMOKE, M129A1, AND YELLOW SMOKE, M194
PARTS AND LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-S-13303G (AR), dated 14 September 1984, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1, Under Standards:

Add COMMERCIAL

ASTM E8 - Tension Testing of Metallic Materials

3.3 Delete in its entirety and substitute following:

"3.3 Cracks. The tube casing shall be free of cracks under visual inspection after hydrostatic testing."

PAGE 3

* 3.4.4e: Delete in its entirety.

* 3.4.4f: Delete in its entirety and substitute the following:

"f. The burning time of the smoke assembly shall not be less than 6.0 seconds nor more than 18 seconds for the M128, and not less than 18.0 seconds nor more than 34 seconds for the M129, and not less than 9.0 seconds nor more than 18 seconds for the M194."

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* 3.4.5b: Delete in its entirety.

* 3.4.5c: Delete in its entirety and substitute the following:

"c. The burning time of the smoke assembly shall not be less than 6.0 seconds nor more than 22 seconds for the M128, and not less than 18.0 seconds nor more than 40 seconds for the M129, and not less than 9.0 seconds nor more than 18 seconds for the M194."

PAGE 9

Table I, Under Assembly, delete "Functioning, minus 65°F, 16(a), 3.4.5, 4.5.2.3.2"

PAGE 13

4.4.2.3 Delay Assembly Add critical defect "1 Delay Charge Missing 100% 3.2 Visual".

Major 101: Delete "Igniter composition above flush" and substitute "Delay Composition above flush or more than max. below flush."

Major 102: Delete in its entirety

PAGE 19

4.4.2.9: Grain Propellant. Add: "Critical 2 Overall Density Test 100% 3.2 4.5.5.1"

Major 103: Delete in its entirety.

Add, under notes:

"a Grain shall be assembled and loaded by same manufacturing methods as intended for production.

b. 25 samples to be tested: accept 0, reject 1.

c. 150 grains to be assembled and tested in 50 signals: accept 0, reject 1."

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PAGE 21

* 4.4.2.11: Smoke Assembly. Delete Major defect 103 in its entirety.

Add Major defect: "104 Quickmatch missing or not partially embedded in the priming charge, 0.40%, 3.2, Visual."

PAGE 25

4.4.2.15: Tail Assembly. Delete in its entirety and substitute the included page.

PAGE 27

4.4.2.17: Tube, Casing. Delete defects in their entirety and substitute the following:

"Critical				
1	Hydrostatic pressure test	100% (Note 1)	3.2	4.5.6
2	Presence of cracks	100%	3.2	Visual
Major				
101	Perpendicularity of ends	0.40%	3.2	Gage
102	Total length, max.	0.40%	3.2	Gage
103	Inside diameter at ends, max.	0.40%	3.2	Gage
Minor				
201	Poor workmanship	1.00%	3.8	Visual

Note 1 - Tubing stock used to manufacture the casing tube shall be 100% hydrostatically tested as specified in Para. 4.5.6. All remaining classified defects shall be checked on the finished item".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Tail Assembly		8797947
			NEXT HIGHER ASSEMBLY 8797996/9255782

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Tail fin missing, cracked or inverted		100%	3.2	Visual
Major					
101	Length, max.		0.40%	3.2	Gage
102	Location of tail vanes with diameter and each other including maximum vane extension from centerline		0.40%	3.2	Gage
103	Inside diameter of upper tail ring, min.		0.40%	3.2	Gage
104	Perpendicularity of ends with diameter		0.40%	3.2	Gage
105	Position of rib inside diameter with upper tail ring diameter		0.40%	3.2	Gage
106	Weld missing		0.40%	3.2	Visual
107	Weld flash excessive to extent that function will be impaired		0.40%	3.2	Visual
108	Weld peel test	(a)	--	3.2	4.5.10
109	Mechanical properties test (See Dwg. 8797951)	(b)	--	3.2	4.5.8
Minor					
201	Clearance between lower tail ring and tail vane		0.65%	3.2	Gage
202	Poor workmanship		1.0%	3.8	Visual

NOTES: (a) 32 samples/lot, 1 upper tail ring weld and 1 lower tail ring weld per assembly (64 weld total); (b) Five(5) tail vanes/test samples shall be selected from each heat treated batch for mechanical properties testing.

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PAGE 28

4.4.2.18: Parachute Assembly: Add to Major 102: “or improperly attached to chain bead assembly, when applicable.”

PAGE 31

4.4.2.21: Assembly (Signal Body & Delay Assembly). Delete Major 101 in its entirety and substitute:

"Major 101 Component missing or damaged to extent that function may be impaired or chain bead assembly improperly fastened to smoke assembly, 0.40%, 3.2, Visual”

PAGE 32

4.4.2.22: Assembly (Prior to Assembly Rocket Assembly, in Rocket Barrel). Add: "Critical 2, Initiating Charge Missing, 100%, 3.2, Visual”

Major 103: Delete in its entirety and substitute: "Major 103, Initiating Charge Over Max. (see note), 0.40%, 3.2, Balance"

PAGE 36

4.4.2.26: Container (Prior to Sealing). Delete Major 103 in its entirety.

PAGE 38A

4.4.2.29: Add new Classification of Defects for Signal Body (Dwg. 8797963) included.

PAGE 39

4.4.3.1: Line 4, delete "Delay Composition (see 3.3)" and substitute "Delay Composition (9251412)”

Line 12, Delete "Fiberboard" and substitute ‘Cardboard”.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.29	Signal Body				8797963
					NEXT HIGHER ASSEMBLY 8797920
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Small inside diameter		0.40%	3.2	Gage
Minor					
201	Large outside diameter, max		0.65%	3.2	Gage
202	Large inside diameter, min		0.65%	3.2	Gage
203	Overall length		0.65%	3.2	Gage
204	Poor workmanship		1.0%	3.8	Visual

NOTES:

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4.4.3.2: Functioning. Add to Table III:

"Cork sealing disc in end of barrel loose Major (see label dwgs. 8797931-2, -3 and -10)"

* Table III: Delete “Smoke emission time over max (See 3.4.4e & 3.4.5b) Major”

Delete NOTE and substitute:

“NOTE: The functioning test shall consist of three(3) distinct phases: Hot temperature - fixture fired; plus 70 degrees - fixture fired; and cold temperature - fixture fired. Reduced sampling will be authorized only when three(3) consecutive lots have met the acceptance criteria specified in all three(3) phases. Compliance with the criteria in any one phase will not result in reduced sampling for that phase.”

4.4.3,2.1.1: First three lots. Line three, delete ‘twenty-five (25)” and substitute "fifty (50)".

4.4.3.2.1.2: After three consecutive lots. Line three, delete "ten (10)” and substitute “twenty (20)”.

4.4.3.2.2.1: First three lots. Line four delete “one hundred and twenty-five (125)" and substitute “one hundred (100)’

Line seven, delete "eight(8)" and substitute "six(6)"

4.4.3.2.3.1.1: First three lots. Line three, delete “thirty-two (32)” and substitute “fifty (50)”

Line four, delete “The lot shall. .(see Table III).” and substitute “The lot shall be rejected if, during the test a critical defect occurs, the average angle from vertical exceeds the requirements, or if four(4) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note after the following paragraph)”.

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4.4.3.2.3,1.2: After three consecutive lots. Line three, delete "ten (10)" and substitute "thirty-two (32)"

Line four, delete "If any critical defect occurs, the lot shall be rejected." and substitute "The lot shall be rejected if during the test, a critical defect occurs, the average angle from the vertical exceeds the requirement or if three (3) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note).

Note: "A smoke assembly which ignites and burns normally but subsequently fails to meet minimum burning time requirements due to ground impact, will not be classified defective. Firing records will be annotated accordingly."

4.4.3.2.3.2: Hand fired: Delete in its entirety.

PAGE 42

4.4.3.3: Propellant Ballistic Test. Delete in its entirety and substitute:

4.4.3.3. Propellant ballistic test - Critical defect. 50 signals per propellant sub-lot of loose-propellant composition are to be ballistically tested in fully loaded signals at hot temperature in accordance with 4.5.2.1 and 4.5.11, Upon successfully meeting the requirements (see 3.4.1) the propellant composition used and the method of manufacture of the propellant assembly shall be considered adequate for production. Al 1 production will then utilize the spacer, propellant part no. 8797936 in the aft end and the slotted spacer propellant part no. 93228587 in the forward (delay assembly end) section of the tube. casing, part no. 8797921."

4.4.3.3.1 Overall density: Delete in its entirety.

4.4.3.3.2: Density of any section. Delete in its entirety.

4*4.3.4: Swivel pull test. Delete in its entirety.

PAGE 43

4.5.1.5: Add new paragraph:

"4.5,1.5 Propellant assembly, propellant grain. The moisture content. of the propellant assembly shall be determined in accordance with the procedure specified in MIL-P-223 or an approved alternate.

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Add the following note after 4.5.2.1, 4.5.2.2 and 4.5.2.3.1:

"Note: "Samples must be fired within three (3) minutes of removal from the conditioning chamber."

PAGE 44

4.5.2.3.2: Hand fired: Delete in its entirety

4.5.6: Hydrostatic test of casing tube. Delete in its entirety and substitute:

"4.5.6 Hydrostatic test of casing tube. In lieu of testing each individual casing tube, the tubing stock, which is used to manufacture the casing tube, shall be 100% tested in a hydrostat test device (see 4.4.4). The minimum pressure shall be applied for not less than 5 seconds. The tubing shall be observed for evidence of failure to comply with the requirements. Any full length of tubing that bursts, cracks, or shows evidence of distortion shall be classified as defective and removed from production".

PAGE 46

* 4.5.8: Hardness of tail vane. Delete in its entirety and substitute:

"4.5.8 Mechanical properties of tail vane. The tail vane/tes samples shall be tested for mechanical properties in accordance with ASTM E8. If any tail vane/test sample fails to comply with the criteria specified on the applicable drawing, the heat-treat batch represented by the sample shall be rejected."

4.5.11: Swivel pull test. Delete in its entirety and substitute:

"4.5.11 Propellant ballistic test. The signal shall use propellant spacer part no. 8797936, in the forward and aft end while following the test procedure of 4.5.2.1. Simulated or dum payloads may be used."

PAGE 47

6.5: Hand firing: Delete in its entirety.

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MIL-S-13303G (AR)
AMENDMENT 4

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